Tech Pocket – E-Commerce Order Management System

1. Entities and Attributes

Entity 1: Customer

Primary Key: Customer_ID

Attributes:

- Customer_ID (PK)
- First_Name
- Last_Name
- Email
- Phone_Number
- Password
- Address
- City
- State
- Pincode

Entity 2: Product

Primary Key: Product_ID

Attributes:

- Product_ID (PK)
- Product_Name
- Category
- Brand
- Price
- Stock_Quantity
- Description
- Warranty
- Image_URL

Examples of Products:

- Mobile Cases
- Chargers
- USB Cables
- Earphones
- Power Banks
- Screen Protectors
- Smartwatches
- Bluetooth Speakers

Entity 3: Category

Primary Key: Category_ID

Attributes:

- Category_ID (PK)
- Category_Name
- Category_Description

Examples:

- Chargers
- Cases
- Audio Accessories
- Power Banks
- Smart Devices
- Screen Protectors

Entity 4: Shopping Cart

Primary Key: Cart_ID

Attributes:

- Cart_ID (PK)
- Customer_ID (FK)
- Product_ID (FK)

- Quantity
- Total_Price

Entity 5: Order

Primary Key: Order_ID

Attributes:

- Order_ID (PK)
- Customer_ID (FK)
- Order_Date
- Total_Amount
- Order_Status
- Delivery_Address

Entity 6: Payment

Primary Key: Payment_ID

Attributes:

- Payment_ID (PK)
- Order_ID (FK)
- Payment_Method
- Payment_Status
- Payment_Date
- Transaction_ID

Entity 7: Inventory

Primary Key: Inventory_ID

Attributes:

- Inventory_ID (PK)
- Product_ID (FK)
- Stock_Available
- Stock_Updated_Date

Entity 8: Admin

Primary Key: Admin_ID

Attributes:

- Admin_ID (PK)
- Admin_Name
- Email
- Password
- Phone_Number

2. Primary Keys

Entity	Primary Key
Customer	Customer_ID
Product	Product_ID
Category	Category_ID
Shopping Cart	Cart_ID
Order	Order_ID
Payment	Payment_ID
Payment_ID	Order_ID
Inventory	Inventory_ID
Admin	Admin_ID

3. Relationships

Entity 1	Relationship	Entity 2
Customer	Places	Order
Customer	Owns	Shopping Cart
Shopping Cart	Contains	Product
Product	Belongs to	Category
Order	Has	Payment
Product	Maintained in	Inventory
Admin	Manages	Product
Admin	Manages	Order
Admin	Updates	Inventory

4. Relationship Cardinality

- **Customer → Order = 1 : M** (One customer can place many orders.)

- **Customer → Shopping Cart** = **1 : 1** (One customer has one active shopping cart.)
- **Shopping Cart → Product** = **M : N** (A cart can contain many products, and a product can appear in many carts.)
- **Category → Product** = **1 : M** (One category contains many products.)
- **Order → Payment** = **1 : 1** (Each order has one payment.)
- **Product → Inventory** = **1 : 1** (Each product has one inventory record.)
- **Admin → Product** = **1 : M** (One admin manages many products.)
- **Admin → Order** = **1 : M** (One admin manages many orders.)
- **Admin → Inventory** = **1 : M** (One admin updates many inventory records.)